

Synchronous Generator (classical)

States: ω, δ

Parameters: H, D, E, V, X, ω_b

Swing equation:

$$2H\dot{\omega} = T_m - T_e - D(\omega - 1)$$

Rotor angle:

$$\dot{\delta} = \omega_b(\omega - 1)$$

Electrical power:

$$P_e = \frac{EV}{X} \sin \delta$$

